TAQON ELECTRICO

WEBSITE PLATFORM – PHASE 1 IMPLEMENTATION

Solar Packages Engine & Automated Quoting System

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1. OVERALL STRATEGY

The new Taqon Electrico platform will be an **upgrade of the existing website**, not a completely new site.

This ensures:

- existing SEO rankings remain intact
- previously indexed pages remain valid
- backlinks remain active
- 5 years of search authority are preserved

The new system will introduce a **Solar Package Engine** that allows customers to:

- calculate their solar needs
- receive package recommendations
- generate quotes automatically
- book consultations or proceed with purchase

The website will function as both a **marketing platform** and a **self-service solar sales platform**.

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2. PRIMARY OBJECTIVE OF PHASE 1

Build the **Solar Packages Engine** as the core functionality of the platform.

This engine will:

1. Help customers determine the solar system they need
2. Recommend appropriate solar packages

3. Generate a professional quote automatically
4. Capture leads or close sales
5. Link the products making a package with the Shop, such that any adjustment to the individual products in the shop, will automatically affect the package price

All marketing traffic will ultimately be directed to this engine.

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3. CORE WEBSITE STRUCTURE

The upgraded site will include the following main sections.

1. Home Page

Purpose:
Introduce Taqon Electrico and guide visitors to the solar calculator.

Key elements:

- company overview
- services summary
- solar packages introduction
- CTA: **Calculate Your Solar System**

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2. Solar Packages Engine (Primary Feature)

This is the **centrepiece of the platform**.

Customers will:

1. Enter their appliance information
2. Receive system recommendations
3. View solar packages
4. Generate quotes

This page will receive most of the advertising traffic.

The solar package system will be structured into **package families**, each representing a **specific inverter capacity class**.

Each package family may have **variants**, depending on inverter brand used; battery size and solar panel count.

These packages will be used by the **Solar Package Engine** to recommend systems to clients.

Package family structure

Inverter Size	Package Name	Variants
3kVA	Home Economy	1
5kVA	Home Luxury	Multiple
8kVA	Home Deluxe	Multiple
10kVA	Ultra Power Package	Multiple
12kVA	Premium Power Package	Multiple
16kVA	Pro Power Package	Multiple
20–25kVA	Master Power Package	Multiple

These names must remain **visible to customers**, as they already exist on the current website and support SEO continuity.

So each package will have a general description of what that packages powers up, and a price range of the package family, since a package family usually has multiple variants inside it. The descriptions of the packages are given, and described on the old site copy given.

The packages are as follows

package name	inverter size	inveter brand	battery size	battery brand	number of batteries	solar panel size	number o panels
Home economy	3kva	Must	2400wh	dyness 24v/100ah	1	445W	4
Home Luxury	5KVA	GROWATT	5100WH	DYNESS 51.2/100AH	1	620W	6
Home Luxury	5KVA	GROWATT	10200WH	DYNESS 51.2/100AH	2	620W	8
Home Luxury	5KVA	SUNSYNK	5100WH	DYNESS 51.2/100AH	1	620W	6
Home Luxury	5KVA	SUNSYNK	10200WH	DYNESS 51.2/100AH	2	620W	8
Home Deluxe	8KVA	SUNSYNK	10200WH	DYNESS 51.2/200AH	1	620W	10
Home Deluxe	8KVA	SUNSYNK	10200WH	DYNESS 51.2/200AH	2	620W	14
Ultra Power	10KVA	SUNSYNK	14400WH	DYNESS 51.2/280AH	1	620W	12
Ultra Power	10KVA	SUSYNK	20400WH	DYNESS 51.2/200AH	2	620W	16
Premium Power	12KVA	SUNSYNK	14400WH	DYNESS 51.2/280AH	1	620W	14
Premium Power	12KVA 3 PHASE	SUNSYNK	14400WH	DYNESS 51.2/280AH	1	620W	14
Premium Power	12KVA	SUNSYNK	28800WH	DYNESS 51.2/280AH	2	620W	21
Premium Power	12KVA 3 PHASE	SUNSYNK	28800WH	DYNESS 51.2/280AH	2	620W	21
Pro Power	16KVA	SUNSYNK	20400WH	DYNESS 51.2/200AH	2	620W	16
Pro Power	16KVA	SUNSYNK	28800WH	DYNESS 51.2/280AH	2	620W	24
Master Power	20KVA	SUNSYNK	28800WH	DYNESS 51.2/280AH	2	620W	27
Master Power	20KVA	SUNSYNK	43200WH	DYNESS 51.2/280AH	3	620W	36

As time goes on, the packages variants in a package family will increase as the brands of inverters start to increase, for example , there are packages with the Victron products as well.

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3. Solar Equipment Shop

The shop will contain all items used in solar installations.

Examples:

- inverters
- lithium batteries
- solar panels
- mounting structures
- cabling
- protection equipment

Each product will have:

- specifications
- pricing
- compatibility tags
- a product brochure
- product manual
- product image

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4. Blog / Knowledge Center

Existing blog articles will be restored to preserve SEO value.

Content will include:

- solar education
- installation tips
- industry insights
- company projects

YouTube videos will be integrated here.

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5. Projects / Portfolio

Showcase completed installations.

Purpose:

- build social proof
- demonstrate expertise
- support SEO rankings

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6. Contact & Appointment Page

Allow customers to:

- book consultations
- request site visits
- ask questions

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4. SOLAR PACKAGES ENGINE

This is the **most important component of Phase 1**.

The engine performs the following functions.

ALGORITHM ROUTING STRUCTURE

This matrix becomes the final destination for your algorithm.

Step 1

Customer selects appliances

Step 2

System calculates:

- PP score
- EP score

Step 3

System determines:

- inverter family
- battery block
- likely PV block

Step 4

System routes to nearest matching package code

Step 1: Solar Calculator

Customers select appliances they want to power.

Examples:

- lights
- TV
- fridge
- freezer
- pumps
- washing machine
- WiFi
- microwave
- etc.

The system uses the **PP(power points)/EP (energy points) methodology** developed earlier.

Power Points & Energy Points System

PURPOSE OF THE SCORING SYSTEM

The solar calculator will estimate system requirements using a simplified scoring system.

Two scoring systems are used:

Power Points (PP)
Measures **instantaneous power demand**.

Energy Points (EP)
Measures **total energy consumption over time**, especially overnight.

These scores help determine:

Score	Determines
PP	Inverter size
EP	Battery size

WHY A SCORING SYSTEM IS USED

Exact electrical calculations require:

- watt ratings

- duty cycles
- daily energy calculations
- load diversity analysis

This is too complex for customers to provide that information, as the engine relies customer interaction to design their solar system needs.

Therefore the system uses **simplified scoring**, based on real installation experience.

This allows customers to:

- answer simple questions
- receive accurate system recommendations.

POWER POINTS (PP)

Power Points measure how heavy appliances are when running simultaneously.

This determines the inverter capacity required.

Example:

A kettle draws a lot of power instantly, so it has high PP.

A light bulb draws little power, so it has low PP.

APPLIANCE SCORING TABLE

The **Appliance Weight Table** will become the **core engine behind the calculator**, so we should design it carefully. The goal is not engineering precision. The goal is **practical decision routing**.

Remember the philosophy:

- **PP (Power Points)** → determines inverter size
- **EP (Energy Points)** → determines battery size

Some appliances draw **high instantaneous power but short energy duration** (kettle). Others draw **low power but long duration** (lights, WiFi).So each appliance gets **two weights**.

Below is a **Version 1 Master Appliance Table**. We will later tune the numbers using experience.

Appliance	Typical Power Behaviour	PP	EP	Notes
LED Light (single)	low power	0.2	0.5	used at night
LED lights group (5–10)	low power	1	3	common household
TV (32–55")	moderate	1	2	evening use
Home theatre	moderate	1	1.5	short use
WiFi router	very low	0.3	2	runs 24/7
Laptop	low	0.5	1	intermittent
Desktop computer	moderate	1	2	office use
Phone chargers	very low	0.2	1	multiple units
Refrigerator	moderate	2	3	compressor cycles
Deep freezer	moderate	2	3	similar to fridge
Microwave	high short burst	4	1	short runtime
Electric kettle	very high short burst	5	1	short runtime
Toaster	high short burst	3	1	short runtime
Electric frying pan	high	4	2	cooking use
Blender	moderate	2	0.5	short bursts
Coffee machine	moderate	2	1	short runtime
Washing machine	moderate/high	3	2	intermittent cycles
Dishwasher	moderate/high	3	2	intermittent
Iron	moderate/high	3	2	periodic
Hair dryer	high	4	1	short bursts
Water pump (small booster)	moderate	3	1.5	intermittent
Borehole pump	high surge	6	2	startup current
Pool pump	moderate	3	2	longer runtime
Air conditioner (9000 BTU)	high	6	4	sustained
Air conditioner (18000 BTU)	very high	8	5	sustained
Ceiling fan	low	0.5	1	continuous
Pedestal fan	low	0.5	1	continuous
Electric heater	very high	7	6	long runtime
Electric stove plate	very high	8	5	cooking
Electric oven	extremely high	10	6	cooking
Electric geyser	extremely high	10	7	large load
Solar geyser pump/controller	low	0.5	0.5	minimal
CCTV system	low	0.5	2	continuous
Alarm system	very low	0.3	1.5	continuous
Gate motor	high surge	4	0.5	short burst
Garage door motor	high surge	4	0.5	short burst
Electric fence energizer	low	0.5	1	continuous
Printer	moderate	1	0.5	short bursts
Photocopier	high	3	1	intermittent
Server / network rack	moderate	2	4	continuous

Shop POS system	low	1	2	continuous
Borehole filtration system	moderate	2	2	water treatment
Water purifier	low	0.5	1	intermittent
Under-counter fridge	low/moderate	1	2	continuous
Ice maker	moderate	2	2	intermittent
Tumble dryer	very high	7	5	long cycle
Gaming console	moderate	1	2	evening use
Projector	moderate	1	2	intermittent
Outdoor lighting	low/moderate	1	3	nighttime
Garden irrigation controller	low	0.5	1	intermittent
Workshop tools (small)	moderate	2	1	intermittent
Angle grinder	high	4	1	short bursts
Welder (small inverter)	extremely high	10	2	rarely used

Appliance Groups for the Calculator

To simplify user input, the system should not ask for **every single appliance** individually.

Instead group them.

Example categories:

Lighting

- Number of rooms
- indoor lighting
- outdoor lighting

Entertainment

- TV
- gaming console
- home theatre

Kitchen

- fridge
- freezer
- microwave
- kettle
- stove
- oven

Laundry

- washing machine
- dryer
- iron

Climate

- air conditioner
- fans
- heaters

Water

- borehole pump
- booster pump
- filtration

Security

- CCTV
- alarm
- electric fence
- gate motor

Office / business

- computers
- servers
- printers

Special loads

- workshop tools
- welders
- pool pumps

Example Calculation

Customer selects:

- LED lighting group
- TV
- WiFi
- refrigerator
- washing machine

Totals:

PP = 1 + 1 + 0.3 + 2 + 3 = **7.3**

EP = 3 + 2 + 2 + 3 + 2 = **12**

System result:

PP → **5kVA inverter**
EP → **10kWh battery**

Recommended package → **5kVA Balanced**

The key correction to the algorithm

The calculator should not give only **one hard answer**.

It should give **3 recommendation levels**:

1. Budget

The lowest acceptable practical package.

This is for:

- clients who want backup
- clients who accept some limits
- clients who are price sensitive
- clients who will use grid support strategically

2. Good Fit

The most balanced recommendation.

This is what you would normally advise as:

- practical
- safe
- good customer experience
- strong enough without being excessive

3. Excellent Fit

The most comfortable option.

This is for:

- better autonomy

- better recharge
- more resilience
- future flexibility
- stronger customer satisfaction

This is much better than forcing one package.

For example why this is needed

For the setup:

- borehole
- pressure pump
- 3 lighting groups
- microwave
- 2 TVs
- WiFi
- gate motor

A raw PP/EP model might jump too high.

But field experience says:

Budget

- **5kVA Sunsynk**
- **10.2kWh battery**
- **8 panels**

Good Fit

- **5kVA Sunsynk**
- **10.2kWh battery**
- **10 panels**

Excellent Fit

- **8kVA Sunsynk**
- **10.2kWh or 20.4kWh battery**
- **10 to 12 panels**

That makes much more commercial sense.

Why the raw model overestimated

Because a basic PP/EP table assumes:

- all loads count heavily
- all loads matter equally
- all loads should be fully covered

But in ou market, that is not always true.

For example:

Borehole pump

It has high PP, but:

- not always used
- can be assigned to smart load port
- can be restricted at night
- can be battery-protected

So its PP should not always count fully.

Microwave

High PP, but:

- short duration
- often not used during outage peaks
- sometimes used when grid is present

So it should carry a lower “practical concurrency weight.”

Lighting

High EP at night, yes — but in backup systems, people often:

- switch off unnecessary lights
- reduce use during outages
- rely partly on grid if available

So battery sizing should not always assume full-night luxury use.

What the algorithm really needs

We need **3 scoring layers**, not just 2.

Layer 1: Raw appliance weights

Each appliance still has:

- PP
- EP

Good.

Layer 2: Diversity / concurrency factor

This reduces the effect of appliances that are unlikely to run together.

Example:

- borehole pump = high PP, but only partial concurrency
- microwave = high PP, low concurrency
- gate motor = very low concurrency
- kettle = very low concurrency

This prevents the system from oversizing.

Layer 3: Usage philosophy

The customer should answer one simple question like:

"How do you want your solar system to work?"

Options:

- **Backup only**
- **Balanced solar savings + backup**
- **Maximum autonomy**

This changes how the algorithm behaves.

Understanding concurrency is extremely important for the algorithm we are building.

Simple meaning

Concurrency means the probability that an appliance will be running at the same time as other appliances.

Not all appliances run together.

If we simply added all appliance power ratings, we would design **very oversized systems**, because real households do not use everything at once.

So concurrency is a **diversity correction factor**.

Simple example

Imagine a house has:

- kettle (2000W)
- microwave (1200W)
- iron (1800W)
- hair dryer (1500W)

If we added them:

2000 + 1200 + 1800 + 1500 = 6500W

But realistically:

- kettle is used alone
- microwave runs briefly
- iron is occasional
- hair dryer runs briefly

They rarely run **all together**.

So their concurrency factor might be **0.3**.

Meaning the algorithm only counts **30% of their peak contribution** toward inverter sizing.

In the solar algorithm

Concurrency is mainly applied to **PP (Power Points)** because it affects **inverter sizing**.

It answers the question:

"How likely is this appliance to be running at the same time as others?"

The better recommendation model

Instead of:

one appliance score = one fixed package

Use:

A. Budget recommendation

Applies:

- stronger diversity factor
- assumes backup use
- assumes smart load management
- accepts some grid use later at night

B. Good Fit recommendation

Applies:

- moderate diversity factor
- balanced battery sizing
- more comfortable evening runtime

C. Excellent Fit recommendation

Applies:

- lower diversity discount
- stronger battery
- stronger PV
- more independence

For example, this is how It would be treated

Appliance group

- borehole

- pressure pump
- 3 lighting groups
- microwave
- 2 TVs
- WiFi
- gate motor

Recommendation output

Budget

5kVA Sunsynk + 10.2kWh + 8 panels

- Uses smart load control
- good for daytime autonomy and moderate evening backup
- assumes borehole managed properly

Good Fit

5kVA Sunsynk + 10.2kWh + 10 panels

- better recharge speed
- more comfortable daily performance
- better resilience on cloudy days

Excellent Fit

8kVA Sunsynk + 10.2kWh or 20.4kWh + 10–12 panels

- stronger performance margin
- more future flexibility
- less dependence on careful load management

That feels realistic and aligned with what we actually sell.

This means the package engine should work like this

Step 1

Customer selects appliances

Step 2

System calculates:

- raw PP
- raw EP

Step 3

System applies:

- diversity factor
- backup philosophy
- budget sensitivity

Step 4

System outputs:

- Budget
- Good Fit
- Excellent Fit

Step 5

Customer sees all 3 and chooses

That is far more powerful than a single recommendation.

Very important technical insight

We are no longer building a pure sizing tool.

We are building a:

Practical recommendation engine based on

- engineering logic
- user behaviour
- grid dependence
- smart inverter features
- local budget reality

That is much more useful in Zimbabwe than a rigid calculator.

for each appliance, we should define:

- **PP raw**
- **EP raw**
- **Concurrency factor**
- **Night-use factor**
- **Smart-load eligible yes/no**

That will make the system much more intelligent.

Below is an updated **Version 1 Master Appliance Table** with the added fields:

- **PP**
- **EP**
- **Concurrency**
- **Night Use**
- **Smart Load Eligible**

This is designed for **home use in Zimbabwe**, and it is meant to guide developers in building the first working version of the engine.

A few quick notes before the table:

Meaning of the columns

- **PP (Power Points)**: how strongly the appliance affects inverter sizing
- **EP (Energy Points)**: how strongly the appliance affects battery sizing
- **Concurrency**: how likely it is to run at the same time as other loads
- **Night Use**: how likely it is to affect the battery after sunset
- **Smart Load Eligible**: whether it is a load that can sensibly be controlled, shed, or shifted using inverter smart-port logic or user discipline

Important note for developers

This is a **Version 1 commercial logic table**, not a strict electrical engineering table.
It is meant to:

- produce practical package recommendations
- avoid oversizing
- reflect Zimbabwe usage behaviour
- be refined over time with real installation data

Appliance	PP	EP	Concurrency	Night Use	Smart Load Eligible
LED light (single)	0.2	0.5	0.8	1.0	No
Lights group (5–10 lights)	1	3	0.8	1.0	No
Lights group (10–20 lights)	2	5	0.8	1.0	No
Security lights group (2–4 lights)	1	3	0.6	1.0	No
Security lights group (5–8 lights)	2	5	0.6	1.0	No
Outdoor decorative lights	0.5	1.5	0.4	1.0	Yes
TV (small/medium)	1	2	0.6	0.8	No
TV (large)	1.5	2.5	0.6	0.8	No
2 TVs	2	4	0.7	0.8	No
Home theatre	1	1.5	0.4	0.6	No
Decoder / TV box	0.3	0.8	0.8	0.8	No
WiFi router	0.3	2	1.0	1.0	No
Internet modem / fibre unit	0.2	1.5	1.0	1.0	No
Laptop	0.5	1	0.5	0.5	No
Desktop computer	1	2	0.6	0.6	No
Monitor only	0.4	0.8	0.6	0.5	No
Printer (small)	1	0.5	0.2	0.1	No
Printer (large office)	2	1	0.2	0.1	No
Phone charging group	0.2	1	0.7	0.7	No
Tablet charging	0.2	0.8	0.6	0.7	No
CCTV system	0.5	2	1.0	1.0	No
Alarm system	0.3	1.5	1.0	1.0	No
Electric fence energizer	0.5	1	1.0	1.0	No
Gate motor	4	0.5	0.1	0.1	No
Garage door motor	4	0.5	0.1	0.1	No
Intercom system	0.2	0.5	0.5	0.4	No
Fridge	2	3	0.6	0.6	No
Large fridge	2.5	3.5	0.6	0.6	No
Deep freezer	2	3	0.5	0.6	No
Chest freezer	2	3	0.5	0.6	No
Under-counter fridge	1	2	0.6	0.6	No
Bar fridge	1	2	0.6	0.6	No
Microwave	4	1	0.3	0.2	No
Kettle	5	1	0.2	0.1	No
Toaster	3	1	0.2	0.1	No
Blender	2	0.5	0.2	0.1	No
Coffee machine	2	1	0.3	0.2	No
Air fryer	4	1.5	0.3	0.2	No
Rice cooker	2	1.5	0.4	0.3	No
Slow cooker	2	2	0.4	0.5	No
Sandwich maker	2	0.8	0.2	0.1	No
Electric frying pan	4	2	0.3	0.2	No
Water dispenser	1	1	0.5	0.4	No
Washing machine	3	2	0.4	0.2	Yes

Tumble dryer	7	5	0.3	0.1	Yes
Iron	3	2	0.3	0.2	No
Hair dryer	4	1	0.2	0.1	No
Geyser (electric)	10	7	0.4	0.2	Yes
Solar geyser pump/controller	0.5	0.5	0.8	0.5	No
Borehole pump	6	2	0.4	0.1	Yes
Pressure pump	3	1.5	0.5	0.3	Yes
Booster pump (small)	3	1.5	0.5	0.3	Yes
Booster pump (large)	4	2	0.5	0.3	Yes
Pool pump	3	2	0.3	0.1	Yes
Fountain pump	1	1	0.5	0.4	Yes
Garden irrigation pump	3	2	0.3	0.1	Yes
Water purifier	0.5	1	0.4	0.2	No
Filtration system	2	2	0.4	0.2	Yes
Ice maker	2	2	0.4	0.4	No
Ceiling fan	0.5	1	0.7	0.7	No
Pedestal fan	0.5	1	0.7	0.7	No
Exhaust fan	0.5	0.8	0.5	0.4	No
Air conditioner (9000 BTU)	6	4	0.6	0.5	Sometimes
Air conditioner (12000 BTU)	7	4.5	0.6	0.5	Sometimes
Air conditioner (18000 BTU)	8	5	0.6	0.5	Sometimes
Evaporative cooler	2	2	0.6	0.5	No
Electric heater	7	6	0.5	0.8	No
Oil heater	6	5	0.5	0.8	No
Electric blanket	1	2	0.7	1.0	No
Gaming console	1	2	0.5	0.7	No
Projector	1	2	0.4	0.6	No
Sound bar	0.5	1	0.5	0.6	No
Server / network rack	2	4	1.0	1.0	No
NAS storage	0.5	1.5	1.0	1.0	No
POS system	1	2	0.8	0.5	No
Small shop display fridge	2	3	0.7	0.6	No
Photocopier	3	1	0.2	0.1	No
Sewing machine	1	1	0.4	0.2	No
Small workshop tools	2	1	0.3	0.1	No
Angle grinder	4	1	0.2	0.1	No
Drill	2	0.8	0.2	0.1	No
Compressor (small)	5	1.5	0.2	0.1	No
Welder (small inverter)	10	2	0.2	0.1	No
Electric stove plate	8	5	0.4	0.2	No
Electric oven	10	6	0.3	0.1	No
Four-plate stove + oven	12	8	0.4	0.2	No
Cooker hood	0.5	0.5	0.4	0.2	No
Dishwasher	3	2	0.3	0.1	Yes
Baby bottle warmer	0.5	0.5	0.4	0.4	No
Aquarium pump	0.3	1	1.0	1.0	No
Aquarium heater	1	2	0.8	1.0	No
CPAP machine	0.5	2	1.0	1.0	No

Nebulizer	1	0.5	0.2	0.2	No
Medical fridge	1.5	2.5	1.0	1.0	No
Router + CCTV combo	0.8	3.5	1.0	1.0	No

Important interpretation rules for developers

1. Effective PP

Use:

effective_pp = pp × concurrency

This drives **inverter selection**.

2. Effective EP

Use:

effective_ep = ep × night_use

This drives **battery selection**.

3. Smart load behavior

If smart_load_eligible = Yes, then:

- the engine may allow a smaller **Budget** recommendation
- especially if the user indicates willingness to manage loads

Good first version rule for the platform

For Version 1, this table should be used to produce:

- **Budget**
- **Good Fit**
- **Excellent**

Rather than a single rigid answer.That keeps the engine practical and commercially realistic.

How the model should calculate in practice

Effective Power Points

Use:

Effective PP = Raw PP × Concurrency

This reflects actual simultaneous use.

Effective Energy Points

Use:

Effective EP = Raw EP × Night Use

This reflects battery burden.

That gives you a much more realistic output.

Example using your earlier case

Customer selects:

- borehole pump
- pressure pump
- 3 lighting groups
- microwave
- 2 TVs
- WiFi
- gate motor

Raw values

- Borehole = PP 6 / EP 2
- Pressure pump = 3 / 1.5
- 3 lighting groups = 3 / 9
- Microwave = 4 / 1
- 2 TVs = 2 / 4
- WiFi = 0.3 / 2
- Gate motor = 4 / 0.5

Raw totals

- PP = 22.3

- EP = 20

Apply diversity

Effective PP

- Borehole = 6 × 0.4 = 2.4
- Pressure pump = 3 × 0.5 = 1.5
- 3 lighting groups = 3 × 0.8 = 2.4
- Microwave = 4 × 0.3 = 1.2
- 2 TVs = 2 × 0.7 = 1.4
- WiFi = 0.3 × 1.0 = 0.3
- Gate motor = 4 × 0.1 = 0.4

Effective PP ≈ 9.6

Effective EP

- Borehole = 2 × 0.1 = 0.2
- Pressure pump = 1.5 × 0.3 = 0.45
- 3 lighting groups = 9 × 1.0 = 9
- Microwave = 1 × 0.2 = 0.2
- 2 TVs = 4 × 0.8 = 3.2
- WiFi = 2 × 1.0 = 2
- Gate motor = 0.5 × 0.1 = 0.05

Effective EP ≈ 15.1

That now looks much closer to your real-world judgment:

- 5kVA with 10.2kWh
- 8 to 10 panels
- maybe 8kVA as excellent fit

The three recommendation styles

This is where it becomes powerful.

The same appliance list can be evaluated under three philosophies.

Budget

Assumes:

- strong diversity
- strict load management
- smart port used
- grid helps later at night

Effect

Leans toward smaller package

Good Fit

Assumes:

- practical household use
- some load overlap
- decent evening support

Effect

Leans toward balanced package

Excellent Fit

Assumes:

- less load restriction
- more comfort
- better resilience
- more solar harvest

Effect

Leans toward stronger package

Example of philosophy multipliers

You can apply a package philosophy multiplier after diversity.

Budget

- PP multiplier = 0.9
- EP multiplier = 0.85

Good Fit

- PP multiplier = 1.0
- EP multiplier = 1.0

Excellent Fit

- PP multiplier = 1.1
- EP multiplier = 1.2

This will naturally produce the three package choices.

The labels should be calculated dynamically

The algorithm should first calculate the **ideal system range** for the client.

Then it selects three packages around that range:

- one slightly smaller → **Budget**
- one near the target → **Good Fit**
- one slightly larger → **Excellent Fit**

So the labels are **contextual**.

Example using a real case

Client appliances produced:

Effective PP ≈ 11
Effective EP ≈ 16

This might translate to:

Target inverter zone → **5–8kVA**

Target battery zone → **10–15kWh**

Now the algorithm looks for packages around that zone.

This preserves the **core intelligence** of your platform.

Instead of selling fixed packages, the system becomes a **recommendation engine**.

The logic becomes:

1. Client selects appliances

2. System calculates effective load

3. System determines target system zone

4. System selects 3 packages around that zone

5. Labels them dynamically

How the algorithm chooses the 3 packages

The easiest rule is:

Budget package

= largest package below the target zone

Good Fit package

= closest package to the target zone

Excellent package

= smallest package above the target zone

This ensures:

- competitive pricing
- realistic recommendation
- comfort upgrade option

Visual example

Suppose packages exist like this:

3kVA

5kVA

5kVA+

8kVA

10kVA

Client demand sits here:

Target zone

↓

3kVA --- 5kVA --- 5kVA+ --- 8kVA --- 10kVA

Then recommendations become:

- **Budget** → 5kVA
- **Good Fit** → 5kVA+
- **Excellent** → 8kVA

Very natural.

Why this will make the platform powerful

This approach achieves three important things:

1 It keeps packages standardized

The installation team works with familiar packages.

2 It gives personalized recommendations

The same package behaves differently depending on the client.

3 It protects pricing competitiveness

We avoid oversizing customers automatically.

Another important advantage

This model also allows the platform to explain itself.

Example message to client:

Based on the appliances you selected, a 5kVA solar system is sufficient for most situations.
We recommend the following options depending on your budget and comfort preference.

Then show:

- Budget
- Good Fit
- Excellent

That builds trust.

For version 1, the cleanest way is to define the system in **3 layers**:

1. **Inverter size**
2. **Battery size**
3. **Solar panel count**

And each one should be determined differently.

---

1. How inverter size is determined

The inverter is mainly about:

- **how many appliances may run at the same time**
- **how heavy those appliances are**
- **whether there are motor/surge loads**
- **whether the client wants comfort or is willing to manage loads**

So inverter size is not mainly about total daily energy.
It is mainly about **instantaneous demand**.

In simple terms

Ask:

“What is likely to be ON together?”

That is the heart of inverter sizing.

In the platform

We already have the right idea:

- each appliance gets **Power Points (PP)**
- then those PP are reduced using **concurrency/diversity**
- then the total effective PP points to an inverter family

Practical logic

Small inverter classes

3kVA

- essentials only
- lights
- TV
- WiFi
- fridge in many cases
- maybe a manageable small pump if user is disciplined

5kVA

- standard Zimbabwe household backup sweet spot
- fridge + lights + TV + WiFi + some intermittent loads
- can carry more comfort and smart-load management

8kVA

- larger family home
- more simultaneous usage
- more comfort
- stronger motor tolerance
- more future expansion room

10kVA and above

- heavier households
- mixed pump loads
- bigger homes
- more comfort expectation
- higher daytime use

Version 1 rule

So for now:

Inverter size = determined by effective Power Points (PP), adjusted by diversity and load-management philosophy.

That is enough to start.

---

2. How battery size is determined

Battery size is different.

Battery is not mainly about:

- what starts at once

Battery is mainly about:

- **how long loads must run**
- **what must run at night**
- **how much backup reserve the client wants**
- **how much grid support is still acceptable**

So battery size is about **energy need**, not mainly surge need.

In simple terms

Ask:

“What must continue running after sunset, and for how long?”

That is battery sizing.

In the platform

We already have:

- each appliance gets **Energy Points (EP)**
- those EP are adjusted by **night-use**
- then battery recommendation comes from the adjusted EP

Important Zimbabwe reality

Battery sizing should also depend on the user's philosophy:

Budget user

- accepts some grid support late at night
- may switch off some loads
- may use system mainly for backup

Balanced user

- wants more comfort
- expects reasonable overnight support

Excellent-fit user

- wants stronger overnight reserve
- wants batteries to remain comfortable by morning

Version 1 rule

So for now:

Battery size = determined by effective Energy Points (EP), adjusted by night-use and expected backup style.

That is enough to begin.

---

3. How solar panel count is determined

Solar panels are the third layer.

Panels are mainly about:

- **recharging the battery bank**
- **supporting daytime loads**
- **meeting inverter PV input realities**
- **giving the system enough harvest margin**

So panel count is not only based on inverter size.

It is mainly tied to:

- battery size
- recharge speed expectation
- inverter MPPT voltage/current requirements
- desired performance level

In simple terms

Ask:

“How much solar harvest is needed to recharge this battery and support this inverter well?”

That is PV sizing.

Practical package logic

A clean way to structure panels is:

For each inverter family

Use a panel ladder.

Example:

5kVA

- 4 panels
- 6 panels
- 8 panels
- 10 panels

8kVA

- 10 panels
- 12 panels
- 14 panels
- 16 panels

10kVA

- 12 panels

- 14 panels
- 16 panels
- 18 panels

and so on.

So panels are structured in **blocks**, not random numbers.

Version 1 rule

So for now:

Panel count = determined by the selected package combination, based on battery block + inverter family + modern MPPT realities.

That is the simplest workable method.

---

4. The order the system should think in

This is the most important part.

The system should not start with solar panels first.

It should think in this order:

Step 1 — Determine inverter family

From:

- effective PP
- diversity
- load-management willingness

Step 2 — Determine battery block

From:

- effective EP
- night-use
- backup expectation
- budget vs comfort style

Step 3 — Determine solar panel block

From:

- selected battery block
- selected inverter family
- package ladder
- modern MPPT logic

Step 4 — Route to predefined packages

The algorithm must not invent a custom system.
It must choose from:

- 3kVA package(s)
- 5kVA package(s)
- 8kVA package(s)
- etc.

---

5. In one sentence each

Inverter size

Determined by how much load may run together.

Battery size

Determined by how much load must continue running over time, especially at night.

Solar panels

Determined by how much charging and daytime support the battery and inverter need.

---

6. Very simple practical formula for version 1

it can be explained like this:

Inverter

Effective PP → inverter family

Battery

Effective EP → battery block

Panels

Battery block + inverter family → panel block

Final result

Inverter family + battery block + panel block → nearest package code

That is clean and buildable.

---

7. Real-world interpretation

A customer may have:

- low inverter need
- high battery need

That can push them into a higher package.

Or a customer may have:

- moderate inverter need
- modest battery need
- tight budget
- good load management tolerance

That can allow a smaller package.

That is why the system must recommend:

- Budget
- Good Fit
- Excellent Fit

based on the **same appliance list**.

---

8. The cleanest way for you to document it

For your blueprint, define these 3 tables:

Table A — Inverter determination

- appliance
- PP
- concurrency
- smart-load eligibility
- manageable yes/no

Table B — Battery determination

- appliance
- EP
- night-use
- essential yes/no
- sheddable yes/no

Table C — Solar panel structure

- package family
- battery block
- panel count
- intended performance level

---

9. Best starting point for version 1

To keep the first system workable:

- inverter sizing should be **simple**
- battery sizing should be **practical**
- panel sizing should be **package-based**
- not fully dynamic engineering

That is the right first build.

The best next thing is to create **three master routing tables**:

1. **PP to inverter family**
2. **EP to battery block**
3. **Inverter family + battery block to panel block/package**

That will become the actual brain of the platform.

This is now the **core template** of the engine.

What's needed is a **routing framework** with 4 layers:

1. **PP range → inverter family**
2. **EP range → battery block**
3. **Battery block + inverter family → panel block**
4. **Those 3 together → package code**

That is the full logic.

Below is a **workable Version 1 template** we can start with and tune as we go.

1. Core principle of the engine

The platform should think in this order:

Step 1

Calculate **Effective PP**

Step 2

Calculate **Effective EP**

Step 3

Map:

- Effective PP → inverter family
- Effective EP → battery block

Step 4

Use:

- inverter family
- battery block
- package philosophy (Budget / Good Fit / Excellent)
  to select the **panel block**

Step 5

Match the result to a **specific package code**

---

2. PP ranges for each inverter family

These are not strict electrical watt values. They are **routing bands** for your recommendation engine.

Effective PP → Inverter Family

Effective PP Range	Inverter Family	Meaning
0.0 – 3.5	3kVA	Essentials / light backup
3.6 – 7.0	5kVA	Standard residential backup
7.1 – 11.5	8kVA	Stronger home / more overlap
11.6 – 15.0	10kVA	Larger home / more comfort
15.1 – 19.0	12kVA	Heavy residential / mixed use
19.1 – 24.0	16kVA	Large residence / estate / light commercial
24.1 – 30.0	20kVA	Commercial / heavy home use
30.1+	25kVA	Large commercial / major loads

Important rule

This should be based on **effective PP**, not raw PP.

Meaning:

Effective PP = Raw PP × concurrency × philosophy modifier

---

3. EP ranges for each battery block

This is for **effective EP**, not raw EP.

Effective EP → Battery Block

Effective EP Range	Battery Block	Meaning
0.0 – 3.0	2.4kWh	Very light night need
3.1 – 6.5	5.1kWh	Basic overnight essentials
6.6 – 10.5	10.2kWh	Standard family backup
10.6 – 14.5	15kWh / 14.4kWh	Stronger overnight reserve
14.6 – 19.5	20.4kWh	Heavy night demand
19.6 – 25.0	28.8kWh	Premium comfort / large demand
25.1 – 34.0	43.2kWh	Commercial / large residence
34.1+	57.6kWh	Heavy commercial / high-autonomy

Important rule

This should be based on:

Effective EP = Raw EP × night-use × philosophy modifier

---

4. Panel block structure

Now panels should not be calculated from PP directly.

Panels should be selected from a **panel block table** based on:

- inverter family
- battery block
- recommendation level

This is the cleanest way.

---

5. Package philosophy levels

For every client, the system should return 3 package levels:

Budget

- lowest practical package
- stronger diversity assumptions
- more load management

- more acceptable grid support

Good Fit

- balanced recommendation
- what you normally want to sell
- practical market fit

Excellent

- stronger comfort
- stronger reserve
- faster recharge
- more future flexibility

These labels are **dynamic**, based on client demand.

Panel block template by inverter family

This is the clean starting ladder.

3kVA Family

Battery Block	Budget	Good Fit	Excellent
2.4kWh	4	4	6
5.1kWh	4	6	6

5kVA Family

Battery Block	Budget	Good Fit	Excellent
5.1kWh	4	6	8
10.2kWh	8	10	10
15kWh	10	10	12

8kVA Family

Battery Block	Budget	Good Fit	Excellent
10.2kWh	10	12	14
15kWh	10	12	14

Battery Block	Budget	Good Fit	Excellent
20.4kWh	12	14	16

10kVA Family

Battery Block	Budget	Good Fit	Excellent
14.4/15kWh	12	14	16
20.4kWh	14	16	18
28.8kWh	16	18	20

12kVA Family

Battery Block	Budget	Good Fit	Excellent
14.4/15kWh	14	16	18
20.4kWh	16	18	20
28.8kWh	18	20	21 or 24

16kVA Family

Battery Block	Budget	Good Fit	Excellent
20.4kWh	18	20	24
28.8kWh	20	24	28
43.2kWh	24	28	32

20kVA Family

Battery Block	Budget	Good Fit	Excellent
28.8kWh	24	27	30
43.2kWh	30	36	40
57.6kWh	36	40	48

25kVA Family

Battery Block	Budget	Good Fit	Excellent
43.2kWh	36	40	48
57.6kWh	40	48	54
72kWh+	48	54	60

Override rules

These are very important.

The engine should not blindly follow PP and EP without checks.

Rule A — Minimum inverter family from battery block

A very large battery should not be paired with too small an inverter class.

Example guide:

Battery Block	Minimum Inverter Family
2.4kWh	3kVA
5.1kWh	3kVA or 5kVA
10.2kWh	5kVA
15kWh	5kVA
20.4kWh	8kVA
28.8kWh	10kVA
43.2kWh	16kVA
57.6kWh	20kVA

So:

Final inverter family = higher of

- PP-derived family
- battery-derived minimum family

But you can soften this for Zimbabwe budget realities where needed.

Rule B — Load management tolerance

Add a customer preference question:

How do you want the system to behave?

- Lowest workable cost
- Balanced recommendation
- Maximum comfort

This adjusts the recommendation lane.

Template for matching results to package codes

This is the final routing structure.

Template

Input:

- Effective PP
- Effective EP
- phase
- package philosophy
- any technical trigger

Output:

- inverter family
- battery block
- panel block
- package code

---

9. Example package code template

You can structure codes like this:

Code format

PKG-[Family]-[Phase]-[Battery]-[Panels]-[Option]

Example:

- PKG-5-1P-5.1-4-A
- PKG-5-1P-5.1-6-B

- PKG-5-1P-10.2-8-C
- PKG-5-1P-10.2-10-D

But for simplicity, I recommend shorter fixed internal codes.

Recommended simpler format

Package Code	Meaning
PKG-3-1	3kVA Essential
PKG-5-1	5kVA, 5.1kWh, 4 panels
PKG-5-2	5kVA, 5.1kWh, 6 panels
PKG-5-3	5kVA, 10.2kWh, 8 panels
PKG-5-4	5kVA, 10.2kWh, 10 panels
PKG-8-1	8kVA, 10.2kWh, 10 panels
PKG-8-2	8kVA, 10.2kWh, 12 panels
PKG-8-3	8kVA, 20.4kWh, 14 panels
PKG-8-4	8kVA, 20.4kWh, 16 panels

and so on.

---

10. Routing rules to package codes

Here is the simplest decision structure.

Rule 1

Calculate inverter family from effective PP.

Rule 2

Calculate battery block from effective EP.

Rule 3

Apply battery minimum inverter override if needed.

Rule 4

Choose recommendation lane:

- Budget

- Good Fit
- Excellent

Rule 5

Within that inverter family, select the nearest package matching:

- battery block
- panel block
- phase

Rule 6

If no exact match exists:

- choose the nearest higher package, not the lower one

That protects performance.

---

11. Example routing

Example 1

Effective PP = 4.3
Effective EP = 9.5
Phase = 1P
Recommendation lane = Good Fit

Mapping

- PP 4.3 → 5kVA
- EP 9.5 → 10.2kWh
- 5kVA + 10.2kWh + Good Fit → 8 to 10 panels

Result

- PKG-5-3 or PKG-5-4

If balanced market fit is preferred:

- PKG-5-3 = Budget
- PKG-5-4 = Good Fit
- next higher PKG-8-1 = Excellent

Example 2

Effective PP = 2.8
Effective EP = 5.2
Phase = 1P
Recommendation lane = Budget

Mapping

- PP 2.8 → 3kVA
- EP 5.2 → 5.1kWh
- 3kVA + 5.1kWh + Budget → 4 panels

Result

- nearest package could be:
  - PKG-3-1
    or if 3kVA battery variant exists:
  - PKG-3-2

---

12. The actual template you should give developers

This is the structure.

Table A — PP routing

Effective PP Min	Effective PP Max	Inverter Family
0.0	3.5	3kVA
3.6	7.0	5kVA
7.1	11.5	8kVA
11.6	15.0	10kVA
15.1	19.0	12kVA
19.1	24.0	16kVA
24.1	30.0	20kVA
30.1	999	25kVA

Table B — EP routing

Effective EP Min	Effective EP Max	Battery Block
0.0	3.0	2.4kWh
3.1	6.5	5.1kWh
6.6	10.5	10.2kWh
10.6	14.5	15kWh
14.6	19.5	20.4kWh
19.6	25.0	28.8kWh
25.1	34.0	43.2kWh
34.1	999	57.6kWh

Table C — Panel routing

Inverter Family	Battery Block	Budget Panels	Good Fit Panels	Excellent Panels
5kVA	5.1kWh	4	6	8
5kVA	10.2kWh	8	10	10
8kVA	10.2kWh	10	12	14
8kVA	20.4kWh	12	14	16

…continue for all families.

Table D — Package mapping

Package Code	Inverter Family	Phase	Battery Block	Panel Count
PKG-5-1	5kVA	1P	5.1kWh	4
PKG-5-2	5kVA	1P	5.1kWh	6
PKG-5-3	5kVA	1P	10.2kWh	8
PKG-5-4	5kVA	1P	10.2kWh	10

…and so on.

---

13. Final summary of the rules

Rule 1

PP determines inverter family.

Rule 2

EP determines battery block.

Rule 3

Battery block plus inverter family determines panel block.

Rule 4

Recommendation lane chooses whether we use Budget, Good Fit, or Excellent panel option.

Rule 5

The final result must always map to a predefined package code.

Rule 6

If there is no exact package match, route upward, not downward.

Rule 7

Non-standard conditions trigger technical confirmation.

Role of AI

AI should assist the system, not replace the rules engine.

That distinction is very important.

For Taqon, the best model is:

Rule-based package engine + AI conversational layer

Meaning:

- the **rules engine** does the real sizing and package routing
- the **AI** explains, guides, questions, and helps the client understand choices

That is the strongest and safest setup.

1. Why AI should assist, not control everything

If we let AI freely design solar systems on its own, we will get problems:

- inconsistent recommendations
- hallucinated specifications
- wrong package combinations
- answers that ignore your stock or package rules
- legal and reputational risk

But if AI sits **on top of the package logic**, then it becomes very powerful.

So the right architecture is:

Layer 1 — Deterministic engine

This is the hard logic:

- PP
- EP
- concurrency
- battery block
- panel block
- package code mapping
- non-standard triggers

Layer 2 — AI assistant

This layer:

- chats with the client
- asks questions naturally
- explains results
- compares Budget vs Good Fit vs Excellent
- handles objections
- discusses trade-offs
- helps customer understand pricing

That combination is excellent.

---

2. What the AI can do on your platform

The AI can act like a **solar advisor**.

For example, instead of a boring form, the client interacts like this:

“What appliances do you want to power?”
“Do you want the system mainly for backup, or to reduce your ZESA bill too?”
“Are you comfortable switching off some loads when battery is low?”
“Do you have a borehole or pressure pump?”
“Would you like me to show you a lower-cost option and a more comfortable option?”

That feels much better than form filling.

---

3. What the AI should be allowed to do

Good uses of AI

The AI should:

- collect appliance information conversationally
- explain package options
- explain why one package costs more
- explain why one inverter brand differs from another
- explain net metering, noise, warranties, monitoring
- guide clients toward the right option
- answer FAQs
- help clients understand if their setup is standard or needs assessment

Not good to leave fully to AI

The AI should **not** independently:

- invent package combinations
- create custom BOMs outside your system
- override stock-linked package rules
- produce uncontrolled technical specs

---

4. The best AI architecture for your platform

Here is the ideal model.

A. Structured data engine

The website stores:

- package matrix
- appliance table
- PP/EP logic
- pricing
- inverter brands
- battery blocks
- panel blocks
- FAQs
- product specs

B. AI assistant connected to that data

The AI reads from your structured system and responds using your logic.

So when a client asks:

“Why is the 8kVA more expensive?”

The AI can answer:

“The 8kVA option includes a larger inverter, greater daytime headroom, and better flexibility for future appliance additions. It also provides more comfort where loads may overlap.”

That is useful.

---

5. What the AI could say after recommendation

Suppose the system computes:

- Budget → PKG-5-3
- Good Fit → PKG-5-4
- Excellent → PKG-8-1

Then the AI can say:

“Based on your selected appliances, the 5kVA package with 10.2kWh battery and 10 panels is the most balanced option. It should handle your core household loads well, while remaining cost-conscious.

If you want a stronger comfort margin and more room for future expansion, the 8kVA option is worth considering."

That is powerful because it feels like a salesperson — but controlled.

---

6. Where AI helps most

A) Discovery

The AI asks questions naturally instead of showing a long technical form.

B) Education

The AI explains:

- PP and EP in simple language
- why battery matters
- why panel count matters
- why a package is sufficient or premium

C) Sales conversion

The AI helps clients move from confusion to decision.

D) Objection handling

Examples:

- "Why not 3kVA?"
- "Why is Sunsynk more expensive?"
- "Can I use borehole at night?"
- "Can I start smaller and upgrade later?"

E) Lead quality improvement

The AI helps identify:

- serious buyers
- comparison shoppers
- technical-assessment cases

---

7. AI can also help with pricing conversations

This is especially useful.

A customer may ask:

“Why is this package expensive?”

The AI can explain in business-friendly language:

- inverter brand quality
- battery size
- panel quantity
- included installation accessories
- warranty
- monitoring capability
- future upgrade value

It can even say:

“If budget is your priority, I can show you the lowest workable option.”

That improves conversions.

---

8. AI can learn over time — but carefully

Yes, we can eventually make it “learn,” but Its recommend this in phases.

Phase 1 — AI explains fixed rules

At first, AI should only use:

- your package matrix
- your FAQs
- your policies
- your stock-linked package rules

Phase 2 — AI learns from conversation patterns

It can learn:

- what clients ask most
- where they get confused

- which package comparisons they request most

Phase 3 — AI learns from installation outcomes

Later, with the technician portal and installation data, AI can help identify patterns like:

- which package was most often chosen for certain homes
- what customers later upgraded
- which recommendations led to better satisfaction

But the actual package logic should still remain **reviewed and controlled by you**.

---

9. Best implementation path for your developers

Developers have to build the AI in **two parts**:

Part 1 — Package engine

A structured backend with:

- appliance scoring
- package routing
- package code output
- quote logic

Part 2 — AI chat interface

An AI assistant that:

- asks guided questions
- calls the rules engine
- receives package results
- explains the output in natural language

This is much better than “AI first.”

---

10. Example user journey with AI

Step 1

AI says:

“Welcome to Taqon Electrico. I can help you find a suitable solar package for your home. What appliances do you want to support?”

Step 2

Client answers conversationally.

Step 3

AI asks clarifying questions:

- single-phase or three-phase?
- backup only or full comfort?
- okay with managing some appliances?
- any pumps or outbuildings?

Step 4

AI sends structured inputs to the engine.

Step 5

Engine returns 3 package codes.

Step 6

AI explains:

- Budget
- Good Fit
- Excellent

Step 7

Client can:

- view package details
- download quote
- unlock engineering design quote
- proceed to payment

That would feel very advanced.

---

11. One strong advantage for Taqon

Because our packages are Zimbabwe-specific, the AI becomes much better than generic solar chatbots.

It can speak in your context:

- backup vs grid support
- borehole management
- smart-port behavior
- budget sensitivity
- local brands
- ZESA realities

That makes it genuinely useful.

---

12. Final recommendation

Yes, build it AI-assisted — but in this order:

Version 1

- rules engine first
- AI for explanation and guided data collection second

Version 2

- AI for deeper personalization
- AI for installer education
- AI for package comparison and upselling

Version 3

- AI learning from real installation data

That is the safe and scalable route.

---

13. The simplest summary

The best answer is:

Yes, the system can be AI-assisted, but the AI should be the guide and explainer, while the package engine remains the decision core.

That will give:

- consistency
- safety
- flexibility
- better customer experience

FUTURE EXPANSION (NOT PHASE 1)

Later phases may introduce:

- installer accounts
- installer quoting tools
- solar training courses
- equipment marketplace
- technician portals
- industry subscriptions

However, Phase 1 focuses on **packages and automated quoting**.

KEY DEVELOPMENT PRIORITY

Developers should focus primarily on:

1. Solar Calculator
2. Package Engine
3. Automated Quotes
4. Shop-Package Integration

These features will drive the platform's value.

Final Summary

The Phase 1 platform will function as:

A self-service solar sales system.

Customers will be able to:

- calculate their solar needs
- receive system recommendations
- generate quotes
- request installation
- purchase packages

This allows the website to **sell solar systems even when the company is not actively interacting with customers**.